IBA SMCS School of Mathematics and Computer Science
Leadership • Excellence • Impact

FINAL YEAR PROJECT · INDUSTRY SYMPOSIUM 2026

School of Mathematics and Computer Science

Supervisor: **Abeera Tariq**

folio3

✦ AI × NATURAL LANGUAGE × BUSINESS INTELLIGENCE

IntelliQuery

AI-POWERED CONVERSATIONAL BUSINESS INTELLIGENCE

Talk to Your Database. No SQL Required.

Empowering **non-technical teams** to generate instant, secure insights from enterprise data using **plain English** — zero developer intervention needed.

THE TRANSFORMATION

BEFORE INTELLIQUERY

Data locked behind technical complexity

complex_report.sql

```
SELECT t1.region, t2.tier,
 CASE WHEN t2.tier = 'enterprise'
 THEN SUM(t3.revenue * 1.2)
 FROM orders t1
 INNER JOIN (SELECT …) t2
 LEFT JOIN (… ROW_NUMBER() …) t3
 GROUP BY ROLLUP(yr, qtr, mon)
 HAVING COUNT(*) > 100;
```

⏱ 4+ Hours Wasted Per Report

VS

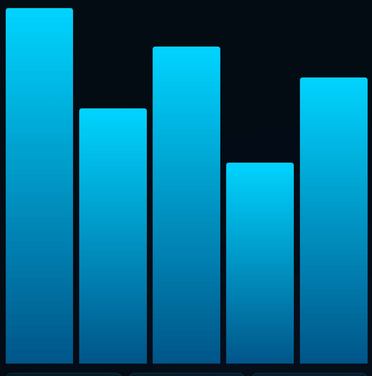
AFTER INTELLIQUERY

Instant insights from plain English

IntelliQuery Dashboard

💬 *"Show revenue by region last month"*

⚡ SQL Generated → Executed → Visualised



$2.4M	+18%	5
Revenue	Growth	Regions

⚡ < 2 Seconds: Question → Insight

99%
SQL Translation Accuracy
LangGraph + Llama 3.3 · 70B via Groq

0
Developer Intervention
End-to-end autonomous pipeline

10×
Faster Data-to-Insight
vs traditional SQL analyst workflow

CORE TECHNOLOGY PILLARS

Llama 3.3 · 70B · LangGraph · LangChain · FastAPI · PostgreSQL · Redis · React 19 · Recharts · ChromaDB · FastEmbed · FastMCP · Tailwind CSS

SUPERVISOR Abeera Tariq

IBA SMCS FYP 2026

TEAM Izma Khan · Sameed Ahmad · Shaikh Muhammad Umair